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ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course:
B.Tech(CSE)
Name of the Paper: Object
Oriented Programming with C++

Semester: III
Paper Code: TCS307

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Discuss basic principles of Object Oriented programming. How classes are different from structure? Explain with suitable example.	
(b)	Define scope resolution operator. Write a program to show two different applications of scope resolution operator in C++.	
(c)	Explain the advantages of using namespace in C++. Write a C++ program that demonstrates how to create a nested namespace Math::Algebra with a function add to add two integer numbers and a nested namespace Math::Geometry with a function area to calculate area of a circle. In main, call both the function and display the results.	
Q2	(20 marks)	CO2
(a)	Discuss the purpose of using different type of constructors in C++. Create a class Height with private members feet and inch. Implement a copy constructor to copy the values from one Height object into another. In the main() function, create a Height object, copy it to another Height object using the copy constructor, and display both Height objects' data.	
(b)	Answer following about static member function of a class with suitable example: (i) Can a static member function be called inside a non-static member function? (ii) What is the relationship between static member functions and this pointer.	
(c)	Differentiate between member function and friend function of a class. There are two classes, String1 and String2, each containing a string as a private member. Write a common friend function mergeStrings() to concatenate the strings from both classes and return the merged result.	
Q3	(20 marks)	CO3
(a)	Discuss the significance of operator overloading in C++ and what are the operators that we can't overload. Write a C++ program to define the class Rectangle with data members length and width, and a method to compute the area. Overload the > operator to compare two Rectangle objects based on the area.	
(b)	Consider the following scenario: - Create a base class Father, having a data member to store father's name, parameterized constructor and a function show () to print the value of data member. - Another base class Mother, having a data member to store mother's name, parameterized constructor and a function show () to print the value of data member.	
	A derived class Child inherits from both the classes Father and Mother, having a data member to store child's name, parameterized constructor and show() function to display the value of data member. Demonstrate how ambiguity occurs when you call show() from the derived class and how to resolve this issue.	

	(c)	Consider a scenario where you have a base class Hospital with <code>getdata()</code> and <code>setdata()</code> methods to store hospital name and address. Two classes Patient and Doctor inherit from Hospital class and having their own data members and member functions to store information of Patients and Doctors. A class Final inherits from both Patient and Doctor class. While implementing this program which kind of problem may arise and how it can be resolved. Explain with code.	CO4
	Q4	(20 marks)	
		Create a class Employee with a method <code>getSalary()</code> . The class has two derived classes: Manager and Developer , both of which override the <code>getSalary()</code> method to provide their specific salary calculation.	
	(a)	- The Employee class should have a member variable <code>name</code> (string) and <code>baseSalary</code> (double). - The Manager class should have an additional member variable <code>bonus</code> (double), and the <code>getSalary()</code> method should return the sum of <code>baseSalary</code> and <code>bonus</code>. - The Developer class should have an additional member variable <code>experienceYears</code> (int), and the <code>getSalary()</code> method should return the <code>baseSalary</code> multiplied by the experience years. In the <code>main()</code> function, demonstrate the above scenario and display the salary of Manager and Developer by calling <code>getSalary()</code> methods.	CO5
	(b)	Explain following with proper example: (i) Virtual Destructor (ii) Pure Virtual Function	
	(c)	Discuss the need of generic programming in modern software development. Write a program to find sum of array elements of any data type using function template.	
	Q5	(20 marks)	
	(a)	Consider a class Demo that has the following member functions: <code>divide(int a, int b)</code>: Divides the first integer by the second and returns the result. Throws an exception if division by zero occurs. <code>subtract(int a, int b)</code>: Subtracts the second integer from the first and returns the result. Throws an exception if the result is negative. Write the C++ code for the class and implement exception handling in the <code>main()</code> function.	CO6
	(b)	Write a C++ program that accepts the following data from the user and writes it to a file named <code>student.txt</code> : - name - roll_number - Marks (3 subjects) Ensure to write the data in a formatted manner in the file. After writing, the program should display a message "the data has been saved". Also read the data from the <code>student.txt</code> file and displays it on the screen in the following format: Name: [name] Roll Number: [roll_number] Marks: [subject1] [subject2] [subject3]	
	(c)	WAP to create an empty vector and perform following operation on it: (i) Insert 5 elements	
		(ii) Delete last element of vector (iii) Delete element from index 2 (iv) Insert an element at a specific position (v) Display size of vector (vi) display elements of vector	